

KUNDA, Estonia

WMO: 260450

Lat: 59.5211N

Lon: 26.5417E

Elev: 14

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.4	3.2	-7.9	3.6	-2.5	-1.8	5.0	4.2	32.5	31.7	28.6	32.1	4.7	160	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	79.2	67.0	75.4	65.6	72.3	64.0	68.9	75.9	66.9	73.4	65.0	71.1	9.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	96.7	73.0	64.3	90.3	70.6	62.4	84.4	69.0	33.1	75.7	31.4	73.6	30.0	71.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.1	24.2	21.1	-6.6	85.0	8.4	4.3	-12.6	88.1	-17.5	90.6	-22.3	93.0	-28.4	96.1 (4)		
(5)			-7.7	71.5	7.8	2.5	-13.3	73.3	-17.9	74.8	-22.2	76.2	-27.9	78.0 (5)		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	24.9	24.0	29.9	39.9	49.2	57.4	63.5	61.5	54.0	44.1	35.4	29.5	(6)
(7)		DBStd	15.56	10.06	10.94	8.29	7.13	7.41	5.40	5.19	5.11	5.62	6.45	7.03	8.92	(7)
(8)		HDD50	3851	777	728	624	315	105	4	0	0	26	201	437	634	(8)
(9)		HDD65	8154	1242	1148	1089	753	493	237	92	136	330	648	887	1099	(9)
(10)		CDD50	1255	0	0	0	12	80	226	418	355	147	17	0	0	(10)
(11)		CDD65	84	0	0	0	0	3	9	45	26	1	0	0	0	(11)
(12)		CDH74	474	0	0	0	0	25	67	242	135	5	0	0	0	(12)
(13)		CDH80	87	0	0	0	0	4	10	46	27	0	0	0	0	(13)
(14)	Wind	WSAvg	9.8	10.9	10.6	10.0	9.0	8.3	9.0	8.3	8.4	9.4	10.8	10.6	11.9	(14)
(15)	Precipitation	PrecAvg	22.8	1.6	1.2	1.1	1.0	1.7	2.6	2.4	3.0	2.2	2.3	2.2	1.5	(15)
(16)		PrecMax	32.4	3.2	3.0	2.2	2.4	4.6	6.7	4.5	7.0	4.0	4.7	3.7	2.5	(16)
(17)		PrecMin	15.6	0.3	0.1	0.3	0.2	0.1	0.6	0.1	0.4	0.7	0.6	0.4	0.4	(17)
(18)		PrecStd	4.1	0.7	0.6	0.6	0.5	1.2	1.5	1.2	1.7	1.1	0.9	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	42.5	50.8	68.2	77.2	81.2	85.7	83.5	72.9	61.0	50.9	48.0	(19)
(20)			MCWB	40.5	39.2	42.3	52.7	63.2	65.3	70.4	66.7	61.7	54.6	48.5	45.0	(20)
(21)		2%	DB	39.4	39.9	45.5	60.3	69.9	74.7	79.6	77.5	68.1	57.4	48.8	44.5	(21)
(22)			MCWB	37.8	37.7	39.4	49.0	57.8	63.8	68.3	65.5	60.1	53.7	46.8	42.5	(22)
(23)		5%	DB	37.6	38.2	41.7	55.0	64.9	70.2	76.1	73.4	65.0	55.2	46.9	42.1	(23)
(24)			MCWB	36.2	36.2	37.5	46.5	55.5	61.4	67.1	64.6	58.6	52.3	44.9	40.1	(24)
(25)		10%	DB	36.0	36.3	39.1	50.6	60.9	66.8	73.2	70.3	62.6	53.1	44.8	39.8	(25)
(26)			MCWB	34.8	34.8	36.0	44.2	52.9	59.2	65.7	63.2	57.6	50.5	43.1	38.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	39.9	43.5	54.0	64.8	68.1	72.1	70.2	63.8	56.4	49.1	45.6	(27)
(28)			MCDB	41.9	41.5	49.6	64.2	74.3	76.6	80.8	77.2	69.0	58.6	50.3	47.6	(28)
(29)		2%	WB	38.0	38.1	40.3	50.3	60.0	65.1	70.1	67.3	61.5	54.4	46.9	42.7	(29)
(30)			MCDB	39.0	39.7	44.5	58.6	67.3	72.4	77.4	73.9	66.5	56.8	48.5	44.4	(30)
(31)		5%	WB	36.5	36.5	38.2	47.4	56.8	62.8	68.2	65.6	59.8	52.6	45.1	40.4	(31)
(32)			MCDB	37.5	38.0	40.9	53.7	62.9	68.9	74.8	72.2	64.1	55.0	46.6	41.9	(32)
(33)		10%	WB	34.9	34.9	36.5	44.7	53.8	60.4	66.3	63.9	58.1	50.6	43.3	38.2	(33)
(34)			MCDB	35.9	36.3	38.7	50.0	59.6	66.0	72.1	69.7	61.8	52.9	44.8	39.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	8.7	10.4	12.8	14.4	13.2	13.3	13.7	11.7	8.6	6.4	6.4	(35)
(36)			MCDBR	6.1	7.6	13.2	20.9	21.2	19.7	18.2	18.4	15.3	10.7	7.8	6.5	(36)
(37)			MCWBR	5.9	6.5	8.8	11.9	11.6	10.8	9.5	9.5	9.0	7.5	7.1	5.9	(37)
(38)		5% WB	MCDBR	6.0	7.4	11.2	19.0	18.9	17.6	16.8	16.1	14.1	9.8	7.5	6.3	(38)
(39)			MCWBR	5.8	6.6	8.0	11.6	11.4	10.7	9.5	9.7	9.4	7.6	7.1	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.308	0.332	0.373	0.370	0.367	0.389	0.387	0.356	0.336	0.310	0.265	(40)	
(41)		taud	2.116	2.134	2.172	2.271	2.389	2.436	2.396	2.399	2.472	2.396	2.206	2.087	(41)	
(42)		Ebn at Noon	158	216	250	262	272	275	265	256	246	210	144	116	(42)	
(43)		Edh at Noon	15	25	32	35	34	33	33	31	25	20	14	10	(43)	
(44)	All-Sky Solar Radiation	RadAvg	78	263	721	1217	1699	1760	1729	1329	809	331	90	46	(44)	
(45)		RadStd	12	51	104	140	179	146	155	136	99	46	15	7	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.21	N/A	N/A	N/A	N/A	N/A	-284	-398	N/A	N/A
(47) Regional (47 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	-274	N/A	N/A	N/A

Nomenclature: See separate page